

Chapter 3. GROUPS

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups



Drawing the lattice diagram of the subgroups of a finite cyclic group

LEMMA 3.31. Let g be an element of a group G of order n . If $d|n$ then the order of the element g^d is n/d .

$$\langle g^m \rangle \subseteq \langle g^k \rangle \Leftrightarrow k|m$$

Example 3.34. Find all subgroups of the multiplicative cyclic group C_{12} and draw its lattice diagram.

Write $C_{12} = \langle g \rangle$, where $o(g) = 12$.

The positive divisors of 12 are

1, 2, 3, 4, 6, 12

By the Fundamental Theorem of Finite Cyclic Groups C_{12} has 6 subgroups of orders 1, 2, 3, 4, 6, 12. Notice that the subgroups of order 1 and 12 are $\{1\}$ and C_{12} , respectively.

• The subgroup of C_{12} of order 2 is of the form $\langle g^d \rangle$, $d = ?$ where $d | 12$

Fundamental Theorem of Finite Cyclic Groups (1)

$$\underline{2} = |\langle g^d \rangle| = o(g^d) = \frac{n}{d} = \underline{\underline{\frac{12}{d}}}$$

↑
Lemma 3.21

$$\Rightarrow d = \frac{12}{2} = 6$$

$$\langle g^6 \rangle = \{1, g^6\}$$

• The subgroup of C_{12} of order 3 ?

It will be of the form $\langle g^d \rangle$, $\underline{d = ?}$

$$3 = |\langle g^d \rangle| = o(g^d) = \frac{12}{d} \Rightarrow \underline{\underline{d = 4}}$$

$$\langle g^4 \rangle = \{1, g^4, g^8\}$$

- The subgroup of C_{12} of order 4

$$\langle g^d \rangle, d = ? \quad (d|12)$$

$$4 = |\langle g^d \rangle| = o(g^d) = \frac{12}{d} \Rightarrow \underline{d=3}$$

$\uparrow$
 Lemma 3.31

$$\langle g^3 \rangle = \{1, g^3, g^6, g^9\}$$

- The subgroup of C_{12} of order 6

$$\langle g^d \rangle, d = ? \quad (d|12)$$

$$6 = |\langle g^d \rangle| = o(g^d) = \frac{12}{d} \Rightarrow \underline{d=2}$$

$\uparrow$
 Lemma 3.31

$$\langle g^2 \rangle = \{1, g^2, g^4, g^6, g^8, g^{10}\}$$

$$\{1\} = \langle 1 \rangle$$

$$\langle g^m \rangle \subseteq \langle g^k \rangle \Leftrightarrow k|m$$

$$\langle g^6 \rangle = \{1, g^6\} \quad \text{order 2}$$

$$\langle g^4 \rangle = \{1, g^4, g^8\}$$

$$\langle g^3 \rangle = \{1, g^3, g^6, g^9\} \quad \text{order 4}$$

$$\langle g^2 \rangle = \{1, g^2, g^4, g^6, g^8, g^{10}\}$$

C_{12}

